

OKTYABR'SKIY GORODOK, Russian Federation

WMO: 341630

Lat: 51.633N

Lon: 45.450E

Elev: 659

StdP: 14.35

Time Zone: 4.00 (E04)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-10.4	-20.9	1.8	-15.3	-15.2	2.5	-10.1	20.4	17.4	18.6	15.9	4.2	80	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	94.0	67.9	90.2	66.6	86.8	65.6	69.7	87.8	68.3	85.8	67.1	83.3	7.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	91.9	74.2	62.5	86.7	73.6	61.1	82.3	73.4	34.0	87.6	32.9	86.1	31.9	83.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.0	16.7	15.1	-23.0	97.6	4.1	3.3	-26.0	100.0	-28.4	101.9	-30.7	103.8	-33.8	106.2		
			-23.3	72.2	4.1	2.0	-26.2	73.6	-28.6	74.8	-30.9	75.9	-33.8	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	12.2	14.5	25.2	45.3	60.6	68.3	72.1	69.5	57.3	42.0	30.9	21.6	(6)
		DBStd	23.13	12.67	12.46	9.80	9.10	7.21	7.85	6.92	7.92	7.31	8.75	9.29	10.55	(7)
		HDD50	4888	1171	994	769	201	4	2	0	17	278	573	879		(8)
		HDD65	8493	1636	1414	1234	593	178	52	16	42	250	712	1022	1344	(9)
		CDD50	2500	0	0	0	58	332	551	686	605	236	31	1	0	(10)
		CDD65	628	0	0	0	0	40	151	237	181	19	0	0	0	(11)
		CDH74	7015	0	0	0	15	435	1612	2652	2091	210	0	0	0	(12)
		CDH80	2861	0	0	0	0	97	628	1181	907	48	0	0	0	(13)
		WSAvg	7.1	7.0	8.1	7.6	7.7	7.0	6.8	6.2	6.1	6.8	7.1	7.1	8.0	(14)
		PrecAvg	16.9	1.3	0.9	0.8	1.0	1.7	1.7	2.0	1.9	1.5	1.3	1.5	1.2	(15)
(16)	Precipitation	PrecMax	27.2	2.8	2.9	2.0	2.4	4.1	4.3	4.3	9.5	3.8	3.6	3.0	2.8	(16)
(17)		PrecMin	8.9	0.1	0.0	0.0	0.1	0.3	0.4	0.1	0.0	0.1	0.0	0.4	0.3	(17)
(18)		PrecStd	4.3	0.7	0.6	0.4	0.6	1.2	1.1	1.1	1.8	1.0	0.9	0.8	0.6	(18)
(19)																
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	34.5	36.6	52.1	76.1	86.5	95.8	98.5	98.6	85.2	68.2	55.2	43.0		(19)
(21)		0.4% MCWB	33.3	34.7	44.6	57.8	62.5	68.8	70.2	68.2	62.4	54.2	51.2	42.0		(20)
(22)		2% DB	33.1	34.8	43.4	71.3	82.6	90.5	94.3	93.2	79.7	63.7	50.6	37.8		(21)
(23)		2% MCWB	32.2	33.3	39.4	55.0	60.8	67.0	67.8	67.5	60.3	52.8	47.9	36.5		(22)
(24)		5% DB	32.0	33.2	39.5	66.6	79.1	87.1	90.7	89.1	75.2	59.1	47.0	35.5		(23)
(25)		5% MCWB	31.5	31.9	36.1	52.8	60.0	66.5	67.4	65.6	58.0	51.2	44.9	34.3		(24)
(26)		10% DB	30.0	31.3	36.6	61.7	74.9	83.3	86.7	85.0	70.8	54.9	43.1	33.6		(25)
(27)		10% MCWB	29.2	30.4	34.1	49.9	58.4	65.4	65.9	64.5	56.4	48.7	41.6	32.6		(26)
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	33.9	35.5	45.0	60.0	65.5	72.2	71.5	70.9	65.2	57.6	51.3	42.3		(27)
(29)		0.4% MCDB	34.3	36.1	50.5	73.5	79.6	88.2	93.6	91.0	80.9	61.2	53.8	42.8		(28)
(30)		2% WB	32.4	33.5	40.3	56.6	63.4	69.8	69.3	68.8	61.6	54.6	48.0	36.7		(29)
(31)		2% MCDB	32.8	34.4	43.4	68.8	78.0	85.9	88.3	88.1	74.6	61.1	50.4	37.5		(30)
(32)		5% WB	31.4	32.1	36.6	53.7	61.2	68.2	68.2	67.3	59.4	51.9	45.1	34.4		(31)
(33)		5% MCDB	32.0	33.0	38.7	64.0	75.5	83.2	86.5	85.5	70.1	58.8	47.1	35.2		(32)
(34)		10% WB	29.3	30.3	34.4	50.7	59.4	66.5	67.1	65.7	57.9	49.5	41.5	32.8		(33)
(35)		10% MCDB	30.0	31.5	36.2	60.8	72.5	79.6	83.8	82.1	67.9	54.4	42.9	33.5		(34)
(36)	Mean Daily Temperature Range	MDBR	11.2	11.3	13.2	18.2	19.5	20.3	21.4	21.8	17.9	13.7	9.0	8.5		(35)
(37)		5% DB MCDBR	9.3	7.9	13.7	26.3	25.3	24.2	26.0	26.1	25.5	19.7	10.7	6.6		(36)
(38)		5% DB MCWBR	9.4	7.5	9.9	13.5	11.1	9.4	9.3	9.6	11.7	11.1	8.8	6.4		(37)
(39)		5% WB MCDBR	9.2	8.0	12.2	25.2	21.6	22.2	23.0	23.5	21.3	17.7	10.5	6.6		(38)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.331	0.364	0.430	0.400	0.388	0.404	0.423	0.376	0.351	0.327	0.300		(40)
(41)		taud	2.107	2.025	2.059	2.094	2.297	2.366	2.319	2.250	2.358	2.365	2.275	2.136		(41)
(42)		Ebn at Noon	215	240	258	254	271	275	268	255	255	238	207	192		(42)
(43)		Edh at Noon	24	35	42	46	39	37	38	39	31	25	21	20		(43)
(44)	All-Sky Solar Radiation	RadAvg	296	644	1073	1376	1813	1948	1933	1634	1112	578	251	174		(44)
(45)		RadStd	67	105	180	124	172	172	159	125	138	87	46	47		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		+1.24	N/A	N/A	+2.25	N/A	N/A	N/A	N/A	+322	+139

Nomenclature: See separate page